• **SUSPENSION—AIR SUSPENSION—DIAGNOSTIC TROUBLE CODES (DTC'S) B2140 OR C2140—RIDE HEIGHT RESET PROCEDURE** • **ELECTRICAL—LAMP—AIR SUSPENSION LAMP ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTC'S) B2140 OR C2140—RIDE HEIGHT RESET PROCEDURE**	**Article No. 01-12-8**

FORD: 1997-2001 EXPEDITION

LINCOLN: 1998-2001 NAVIGATOR

ISSUE

Some vehicles may exhibit a check suspension lamp on with Diagnostic Trouble Codes (DTC's) B2140 or C2140 even after the Air Suspension switch in the upper RH kick panel has been turned on. This may occur at pre-delivery, following suspension service, or after Air Suspension module replacement. This may be caused by the ride height calibration.

ACTION

Adjust and/or recalibrate ride height as necessary using the NGS (New Generation Star) Tester. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE

DURING PDI (PRE-DELIVERY INSPECTION) IF "CK SUSP" LIGHT IS ON, PRIOR TO SENDING TO SERVICE, VERIFY SWITCH IN UPPER RIGHT KICK PANEL IS ON. THIS SWITCH IS TURNED OFF DURING VEHICLE SHIPMENT TO PREVENT MODULE FROM SEEKING TRIM HEIGHT WHILE SECURED DOWN FOR TRANSPORT.

NOTE

AN INADVERTENT B2140 OR C2140 MAY RESULT IF A PARAMETER RESET (CLEAR RIDE HEIGHT) IS SELECTED UNDER NGS ACTIVE COMMANDS, THEN CANCELLED OUT.

NOTE

THE AIR SUSPENSION CONTROL MODULE PART NUMBER IS INTERCHANGEABLE BETWEEN THE REAR AIR SUSPENSION (RAS) AND 4-WHEEL AIR SUSPENSION (4WAS) SYSTEMS, BUT ONCE THE MODULE IS INSTALLED, THE VEHICLE WILL THEN DETERMINE WHETHER VEHICLE IS 4X2 (RAS) OR 4X4 (4WAS) BASED ON VEHICLE HARNESS INPUTS.

NOTE

MODULES ARE PRE-PROGRAMMED TO SEEK 2.66 VOLTS +/- .02 VOLTS FROM REAR HEIGHT SENSOR (RHGSTEN PARAMETER IDENTIFIER OR PID) AT TRIM HEIGHT. HOWEVER, A NEW OR SWAPPED MODULE REQUIRES CALIBRATION PER STEPS BELOW TO CLEAR A 2140 (INITIALIZATION FAILURE) CODE.

NOTE

PERFORM 4WAS-AIR SUSP/EVO (ELECTRONIC VARIABLE ORIFICE STEERING) SELF TEST AND MAKE NECESSARY REPAIRS FOR ANY OTHER AIR SUSPENSION DTC'S THAT ARE PRESENT BEFORE ADDRESSING B2140/C2140.

WARNING

WARNING: BEFORE REPLACING ANY AIR SUSPENSION COMPONENTS, HOISTING, JACKING, OR TOWING, DISCONNECT POWER TO THE SYSTEM BY TURNING OFF THE AIR SUSPENSION SWITCH LOCATED IN THE RIGHT KICK PANEL AREA TO PREVENT VEHICLE DAMAGE OR PERSONAL INJURY.

RAS ONLY - CLEAR 2140

NOTE

EVEN RAS (4X2) EQUIPPED VEHICLES REQUIRE STORING RIDE HEIGHT AND TOGGLING FRONT TO "ON" TO CLEAR A CK SUSP LIGHT/2140 CODE. THIS ALLOWS THE MODULE TO STORE THE VOLTAGE WITH NO FRONT HEIGHT SENSOR PRESENT, AND MUST BE DONE EVEN WITH CONVENTIONAL (STEEL COIL) FRONT SUSPENSION.

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1. Enter 4WAS active commands and trigger on SAVE CAL (Store Ride Height).

NOTE
YOU MUST DISREGARD WARNING THAT FURTHER ACTION PERMANENTLY WRITES TO THE MODULE AND TRIGGER THROUGH.

2. Trigger through the NGS warning message. NGS screen will now split with active commands on the LH side of screen and any PIDS selected on the right.

3. Toggle FRONT to "ON", press Cancel.

NOTE
CANCEL MUST BE SELECTED TO COMPLETE THE OPERATION. DO NOT SAVE OR CLEAR REAR RIDE HEIGHT.

4. Retrieve and clear continuous DTC's; at this point a 2140 code should be present. Press NGS #7 keypad button located directly underneath the word "CLEAR" on the LCD screen, depress trigger per screen prompt, and wait for message stating "system passed - no DTCs available". Earlier NGS software revision levels may require repeating these steps several times before a system pass message is received.

5. Cancel out of continuous test in Step #4 after receiving pass code message. Select on-demand self test, trigger, press NGS #3 keypad button located directly underneath the word "START" on the LCD screen. Note on-screen instructions to prepare vehicle for test/function then trigger to begin the test. You should hear underhood noise from compressor and relays indicating this functional (on-demand) test is in process. As test completes, verify a "system passed" message is received.

6. Cancel out of on-demand test, remove NGS tester, start vehicle and verify the check suspension indicator remains off.

4WAS - CLEAR 2140 DTC - Mechanical

NOTE
OBSERVE AND REPAIR ANY DAMAGED PARTS OR SYSTEM LEAKS, INSPECTING THE AIR LINE UNION LOCATED UNDER THE BRAKE MASTER CYLINDER FIRST. IF BRAKE FLUID HAS DAMAGED THE LINE, REPLACE THEN REROUTE NEW AIR LINE AWAY FROM AN ORIENTATION DIRECTLY UNDER THE MASTER CYLINDER. BRUSH A SOAPY WATER SOLUTION ON REMAINING SYSTEM CONNECTIONS, INCLUDING INLINE AND "T" UNIONS; AIR LINE RUB POINTS; COMPRESSOR FITTINGS; GATE AND FILL SOLENOIDS; AND AIR SHOCKS/SPRINGS. KEEP IN MIND THAT AIR SPRING LEAKS WHERE THE BAG FOLDS OVER ITSELF MAY NOT BE IMMEDIATELY OBVIOUS WITHOUT CYCLING THE SUSPENSION THROUGH DIFFERENT HEIGHTS. IF AIR BUBBLES ARE DETECTED, REPAIR AS NECESSARY THEN RETEST TO ENSURE LEAK-FREE CONNECTIONS BEFORE COMPLETING THE STEPS BELOW.

1. Bring the vehicle to kneel height by turning ignition key off, gear lever in park, doors closed for up to 30 seconds on level ground. Listen for compressor's vent solenoid to release air from lines before proceeding. Venting should occur within approximately one minute of closing all doors with the key in the "off" position.

2. Erase the module's memory:
   - Trigger 4WAS active commands, PARAMETER RESET (Clear Ride Height)

WARNING
DO NOT RESET REAR HEIGHT AT THIS TIME!
   - Trigger thru warning message, then toggle FRONT CLR from "OFF" to "ON"
   - Press Cancel

NOTE
CANCEL MUST BE SELECTED TO COMPLETE THE OPERATION!

3. You have now cleared the front kneel voltage stored in the module and B2140 will be present on next key-up.

4. Bleed air from front shocks:
   - Mark existing torsion bar setting, then back off the torsion bar adjustment bolt several turns (until approximately 40 mm (1.5″) distance from the bolt head to the bottom surface of the torsion bar adjuster is reached)
   - Select AIR SUSP CNTRL from active command menu (NGS will return to this after previous steps)
   - Trigger VENT FRONT, then wait several seconds for air to vent. Press Cancel to exit.

5. Adjust torsion bars per Workshop Manual section 204-01B.

NOTE
LEAVE THE NGS CONNECTED IN ACTIVE COMMAND MODES (ENTERED IN STEP 4) SO THE SYSTEM REMAINS IN DIAGNOSTIC MODE AND NOT OPERATIONAL MODE.

6. Finish by performing the following electrical front height reset steps.

4WAS - CLEAR 2140 DTC - Electrical

1. Enter 4WAS active commands and trigger on SAVE CAL (Store Ride Height).

NOTE
YOU MUST DISREGARD WARNING THAT FURTHER ACTION PERMANENTLY WRITES TO THE MODULE AND TRIGGER THROUGH.

2. Trigger through the NGS warning message. NGS screen will now split with active commands on the LH side of screen and any PIDS selected on the right.

3. Toggle FRONT to "ON", press Cancel.

NOTE
CANCEL MUST BE SELECTED TO COMPLETE THE OPERATION. DO NOT SAVE OR CLEAR REAR RIDE HEIGHT.

4. Retrieve and clear continuous DTC's; at this point a 2140 code should be present. Press NGS #7 keypad button located directly underneath the word "CLEAR" on the LCD screen, depress trigger per screen prompt, and wait for message stating "system passed - no DTCs available". Earlier NGS software revision levels may require repeating these steps several times before a system pass message is received.

5. Cancel out of continuous test in Step 4 after receiving pass code message. Select on-demand self test, trigger, press NGS #3 keypad button located directly underneath the word "START" on the LCD screen. Note on-screen instructions to prepare vehicle for test/function then trigger to begin the test. You should hear underhood noise from compressor and relays indicating this functional on-demand test is in process. As test completes, verify a "system passed" message is received.

6. Cancel out of on-demand test, remove NGS tester, start vehicle and verify the check suspension indicator remains off and vehicle height settles out as compressor fills air bladders/shocks.

NOTE
FOR OTHER AIR SUSPENSION CONCERNS, REFER TO THE APPROPRIATE WORKSHOP MANUAL SECTIONS: 204-05A FOR VEHICLE DYNAMIC SUSPENSION - REAR, OR SECTION 204-05B FOR VEHICLE DYNAMIC SUSPENSION - 4 WHEEL.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
011208A	Recalibrate Ride Height For Rear Air Suspension (R.A.S.) 4X2 Vehicles Only	0.2 Hr.
011208B	Recalibrate Ride Height For 4 Wheel Air Suspension(4W.A.S.) (Includes Time To Adjust Torsion Bars 4X4 Vehicles Only)	0.5 Hr.

DEALER CODING

BASIC PART NO.	CONDITION CODE
5A919	42

OASIS CODES: 206000, 304000, 390000